NAVARASAM ARTS AND SCIENCE COLLEGE FOR WOMEN

ARACHALUR, ERODE – TAMIL NADU

DEPARTMENT OF COMPUTER SCIENCE

B.Sc. COMPUTER SCIENCE

PROJECT DOCUMENTATION

POCKETSMART AI

Your Smart Budget & Recommendation Assistant

Submitted by

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Academic Year: 2026–2027

CERTIFICATE

This is to certify that the project entitled “PocketSmart AI – Your Smart Budget & Recommendation Assistant” is a bonafide project work carried out by the following students of B.Sc. Computer Science, Navarasam Arts and Science College for Women, during the academic year 2026–2027.

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DECLARATION

We hereby declare that the project entitled “PocketSmart AI – Your Smart Budget & Recommendation Assistant” is our original academic project work. The system was designed, implemented and tested by the project team as part of the requirements for B.Sc. Computer Science.

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ACKNOWLEDGEMENT

We express our sincere gratitude to the management, principal, Head of the Department, project guide and faculty members of Navarasam Arts and Science College for Women for their valuable guidance and support. We also thank our friends and family members for their encouragement during the planning, development, testing and documentation of PocketSmart AI.

ABSTRACT

PocketSmart AI is a web-based smart budget and recommendation assistant developed to help users plan purchases and activities according to a specified budget. The system provides three main planning modules: Home Interior Planner, Party Budget Planner and Jewelry Budget Planner. Users can register, log in, enter budget and preference information, receive recommendation plans, view dashboard information and access recommendation history.

The application is implemented using Python, FastAPI, Uvicorn, HTML, CSS and a local database. A recommendation/AI layer can process user preferences and budget constraints to generate practical suggestions. The project demonstrates the integration of web application development, database management and AI-assisted recommendation concepts into a single user-friendly system.

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1. INTRODUCTION

PocketSmart AI is a smart planning web application that helps users make budget-conscious decisions. The user provides a budget and relevant preferences, and the application prepares a recommendation plan for the selected category. The project focuses on home interiors, parties and jewelry purchases.

The application combines a simple browser interface with a Python-based backend, database storage and recommendation logic. It also maintains user-specific history so that previously generated plans can be reviewed later.

2. PROBLEM STATEMENT

Users often spend considerable time comparing alternatives while trying to stay within a fixed budget. Manual searching across different websites, estimating expenses and keeping previous choices in memory can be inconvenient. Therefore, a centralized application is required to collect budget and preference information and generate organized recommendations.

3. OBJECTIVES

- Develop a simple and user-friendly budget planning application.
- Support home interior, party and jewelry planning.
- Accept budget and preference inputs from users.
- Generate practical recommendations based on the supplied information.
- Provide registration and login facilities.
- Maintain recommendation history for later reference.
- Provide a dashboard showing user activity.
- Create a modular system that can be extended with additional planners and live services.

4. SCOPE OF THE PROJECT

The current project covers user authentication, dashboard access, three budget-planning modules and recommendation history. The system is designed as a prototype that can be extended with live product data, external APIs, mobile support and more advanced personalization.

5. EXISTING SYSTEM

In the existing manual approach, users search multiple sources, compare products or services, calculate expenses and make a final decision themselves. This requires repeated effort and does not provide a single personalized history of recommendations.

6. PROPOSED SYSTEM

PocketSmart AI brings budget planning and recommendation features into one application. The user logs in, chooses a planner, provides the required information and receives a recommendation plan. The generated recommendation can then be stored and viewed from the history page.

7. FEASIBILITY STUDY

7.1 Technical Feasibility

The project can be developed using commonly available software such as Python, FastAPI, Uvicorn, HTML/CSS, a local database and VS Code. The development environment shown in the screenshots demonstrates the application running locally.

7.2 Economic Feasibility

The prototype uses standard development tools and does not require expensive hardware. It can therefore be developed with relatively low infrastructure cost.

7.3 Operational Feasibility

The system uses simple forms and navigation, making it suitable for users who are familiar with basic web applications.

8. SYSTEM REQUIREMENTS

8.1 Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM recommended
- Keyboard and mouse
- Internet connection when external AI/online services are used

8.2 Software Requirements

- Windows/Linux/macOS
- Python
- FastAPI
- Uvicorn
- HTML5 and CSS3
- JavaScript where required
- SQLite or configured local database
- Visual Studio Code
- Modern web browser

9. SYSTEM ARCHITECTURE

The application follows a layered web application structure. The browser provides the user interface. FastAPI handles routes and HTTP requests. Application logic validates inputs and coordinates recommendation generation. The database stores user and recommendation information. An AI/recommendation component can process preferences and budget constraints.

Layer	**Responsibility**
Presentation Layer	HTML templates, CSS and browser interface
Application Layer	FastAPI routes, validation and business logic
Recommendation Layer	Budget/preference processing and AI-assisted recommendation
Data Layer	User records and recommendation history
Server	Uvicorn development server

10. SYSTEM FLOW / WORKFLOW

START

↓

Open PocketSmart AI

↓

Register / Login

↓

Dashboard

↓

Select Planner

↓

Home Planner / Party Planner / Jewelry Planner

↓

Enter Budget + Preferences

↓

Validate Input

↓

Generate Recommendation

↓

Display Recommendation

↓

Save to History

↓

END

The workflow ensures that the user moves from authentication to planner selection, provides the necessary information, receives a recommendation and can later retrieve the result from the history module.

11. USE CASE DESCRIPTION

Actor	Use Case	Description
User	Register	Create a new PocketSmart AI account.
User	Login	Authenticate and access personalized features.
User	Dashboard	View account and recommendation activity.
User	Home Planner	Plan home interiors according to budget and style.
User	Party Planner	Plan party expenses according to budget and guest count.
User	Jewelry Planner	Get jewelry ideas according to budget, occasion and style.
User	View History	Review previously generated recommendation plans.

The single primary actor is the user. The use cases represent the main functions visible in the current application interface.

12. MODULES DESCRIPTION

12.1 Home Module

Introduces PocketSmart AI and provides navigation to the application features.

12.2 Registration and Login

Allows users to create an account and authenticate before using personalized features.

12.3 Dashboard

Displays the logged-in user's name and recommendation activity.

12.4 Home Interior Planner

Accepts a total budget, room type and interior style to prepare a home planning recommendation.

12.5 Party Budget Planner

Accepts budget and guest information and supports planning of food, venue, decoration and entertainment.

12.6 Jewelry Budget Planner

Accepts jewelry budget, occasion and style information; the interface also supports optional outfit-image input.

12.7 Recommendation History

Displays earlier generated recommendations along with date/time information.

13. DATABASE DESIGN

The application uses a local database for persistent storage. The core data can be represented using User and Recommendation entities. A user may create multiple recommendation records, giving a one-to-many relationship.

Entity	Important Fields	Purpose
User	user_id, name, email, password	Stores account and authentication information.
Recommendation	recommendation_id, user_id, category, input_data, result, created_at	Stores generated plans and history.

ER Relationship

[USER] 1 ——————— M [RECOMMENDATION]

Relationship: One user can have many recommendation records. Each recommendation belongs to one user.

14. PROJECT FOLDER STRUCTURE

```
PocketSmart_AI/
├── app/
│   ├── __init__.py
│   ├── main.py
│   ├── config.py
│   ├── database.py
│   ├── dependencies.py
│   ├── schemas.py
│   ├── security.py
│   ├── static/
│   │   └── style.css
│   └── templates/
│       ├── base.html
│       ├── home.html
│       ├── index.html
│       ├── login.html
│       ├── register.html
│       ├── dashboard.html
│       ├── history.html
│       ├── party.html
│       └── jewelry.html
├── tests/
│   └── test_app.py
├── pocketsmart.db
├── requirements.txt
└── README.md
```

15. IMPLEMENTATION

The backend is implemented with Python and FastAPI. Uvicorn is used to run the application during development. HTML templates provide the front-end pages and CSS controls the visual appearance. Route handlers provide the home page, login, dashboard, planners and history endpoints. Database and security modules support persistent data and user authentication.

The screenshots show the application running at a local development address and demonstrate successful HTTP requests. The terminal output includes 200 OK responses for major pages and API calls.

16. TESTING AND RESULTS

Test Case	Input / Action	Expected Result	Observed Result	Status
TC01	Open Home page	Home page loads	Page loaded successfully; GET / returned 200 OK	PASS
TC02	Open Login page	Login form displayed	Login page loaded; GET /login returned 200 OK	PASS
TC03	Login with valid details	Dashboard opens	Dashboard displayed; POST /api/login returned 200 OK	PASS
TC04	Open Dashboard	User dashboard displayed	Dashboard loaded; GET /dashboard returned 200 OK	PASS
TC05	Open Home Planner	Home planner form displayed	GET /planner/home returned 200 OK	PASS
TC06	Open Party Planner	Party planner form displayed	GET /planner/party returned 200 OK	PASS
TC07	Open Jewelry Planner	Jewelry planner form displayed	GET /planner/jewelry returned 200 OK	PASS
TC08	Open Recommendation History	Previous recommendations displayed	GET /history and GET /api/history returned 200 OK	PASS

17. SCREENSHOTS

The following screenshots are included as evidence of the implemented user interface and successful local execution.

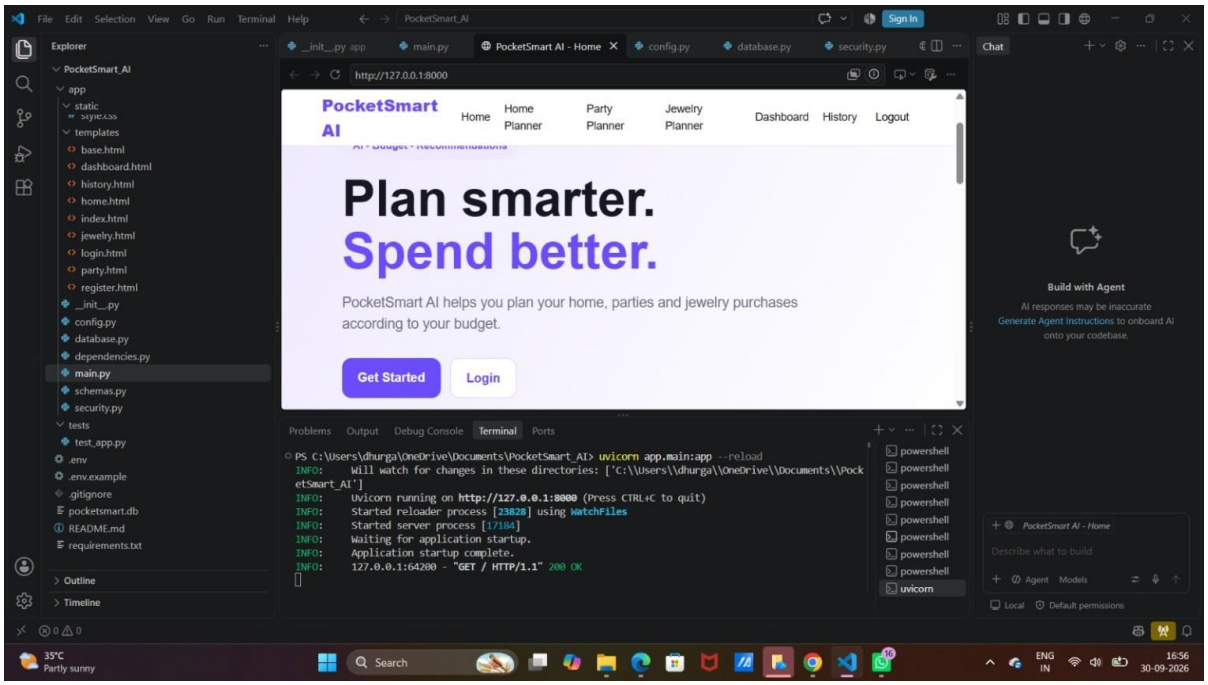


Figure 1: PocketSmart AI Home Page

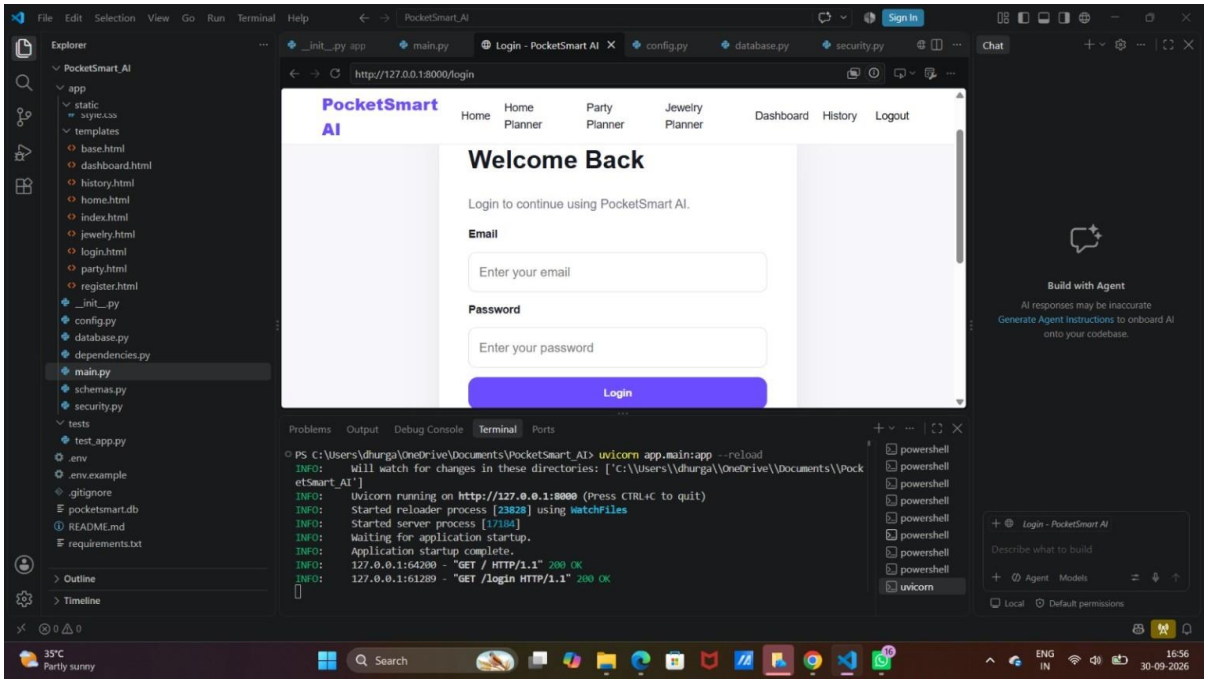


Figure 2: User Login Page

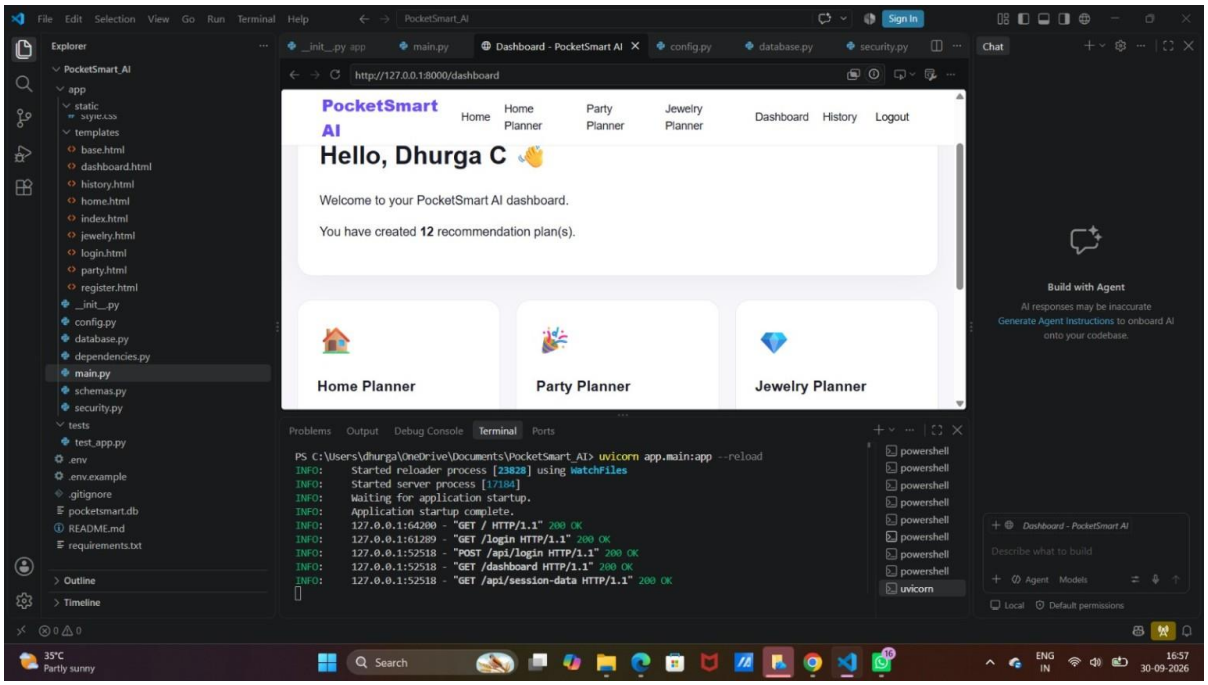


Figure 3: User Dashboard

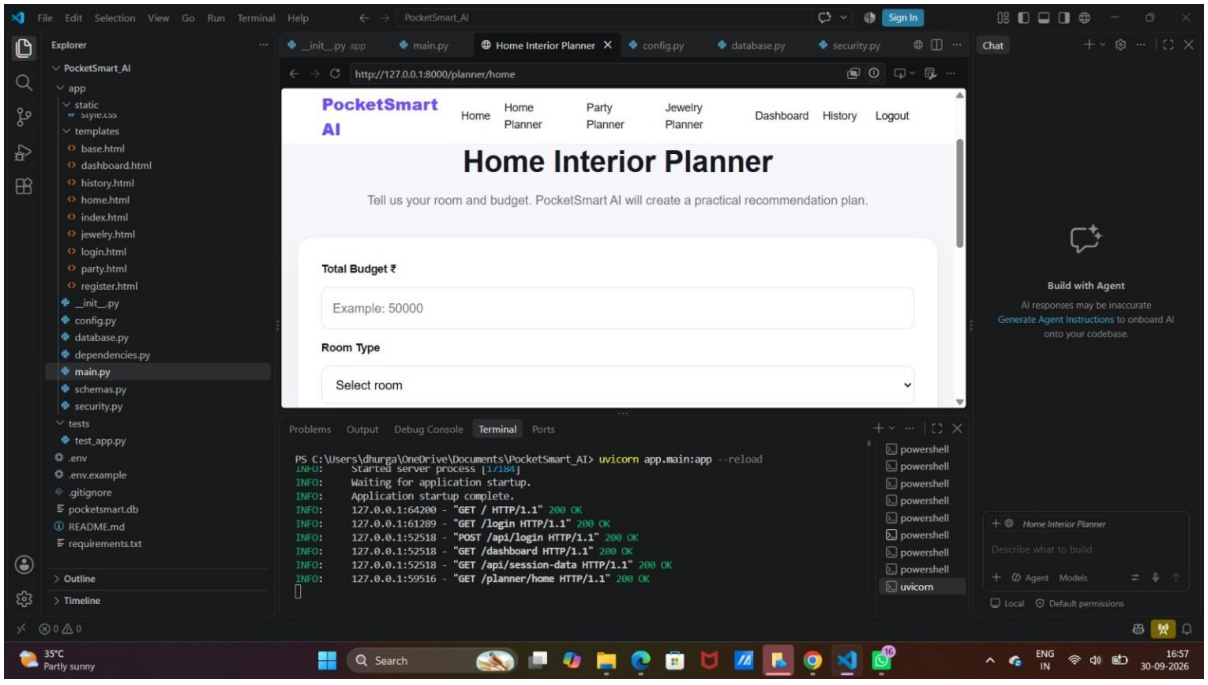


Figure 4: Home Interior Planner

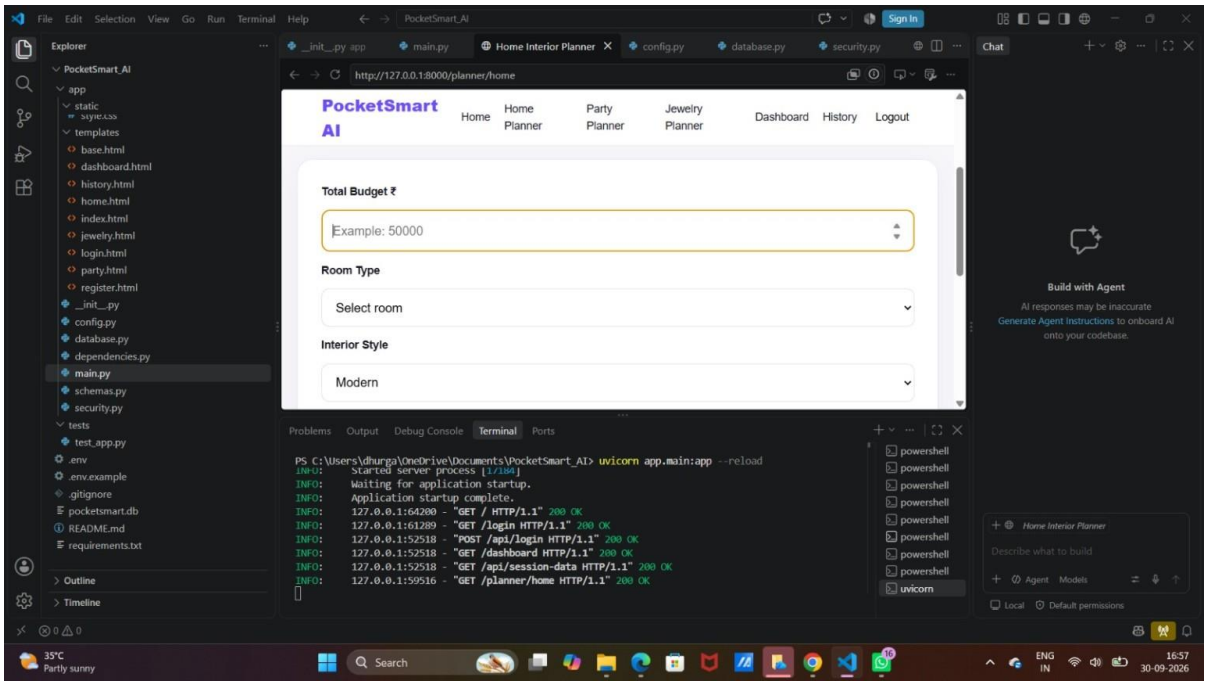


Figure 5: Home Planner – Interior Style Selection

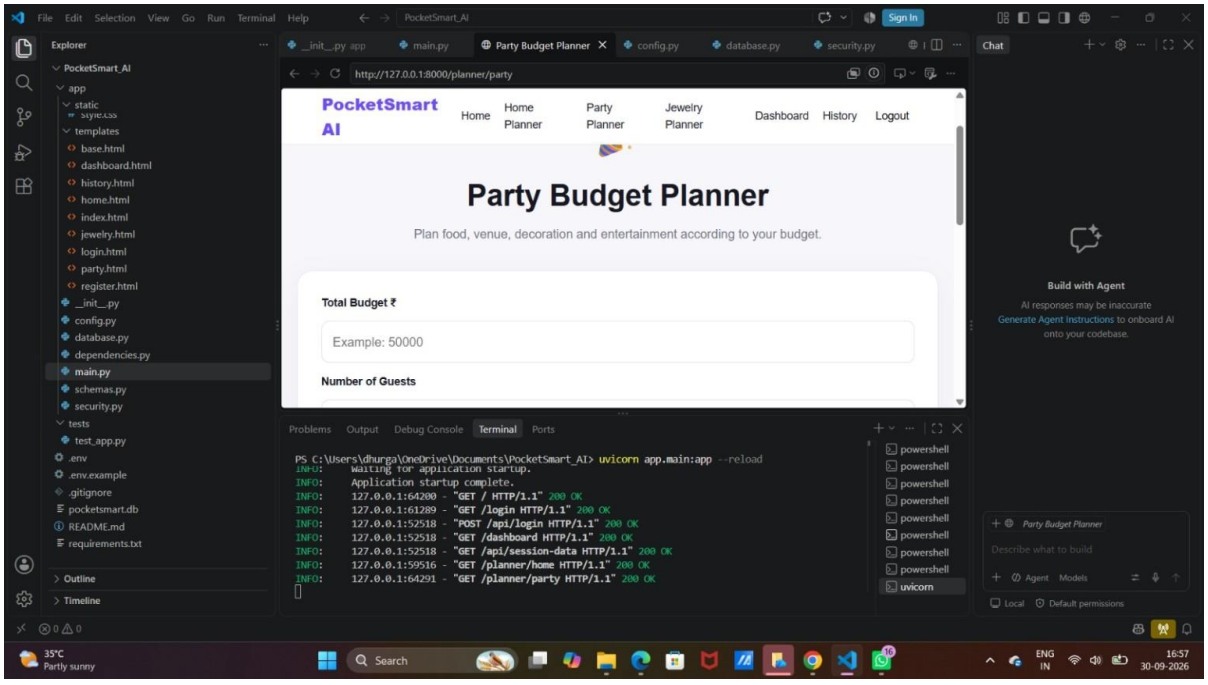


Figure 6: Party Budget Planner

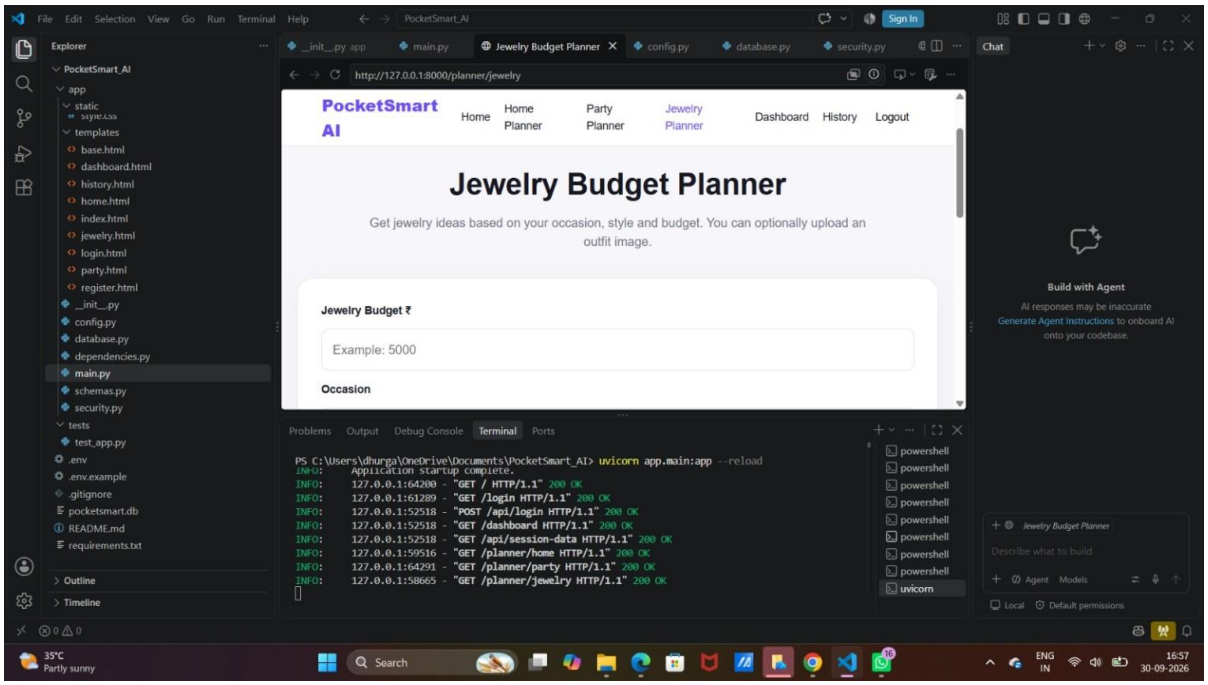


Figure 7: Jewelry Budget Planner

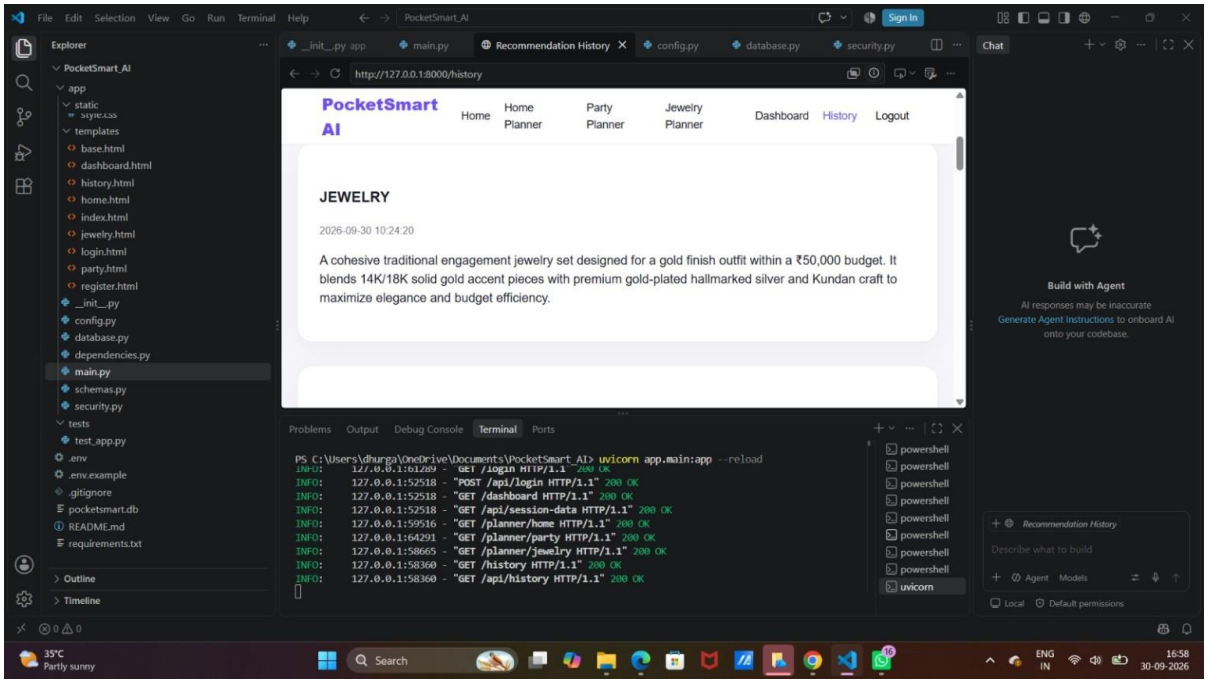


Figure 8: Recommendation History

18. ADVANTAGES

- Simple and user-friendly web interface.

- Budget-oriented planning and recommendations.
- Three separate planning categories in one application.
- Personalized dashboard and recommendation history.
- Modular FastAPI-based architecture.
- Can be extended with live services and additional planners.

19. LIMITATIONS

- Recommendation quality depends on the configured recommendation/AI logic and user input.
- Live prices and availability may change when external sources are used.
- Some external AI or online services may require internet connectivity and API configuration.
- The current implementation is primarily a development/prototype system.

20. FUTURE ENHANCEMENTS

- Integrate live product/service APIs and price comparison.
- Add advanced AI personalization and recommendation feedback.
- Provide mobile application support.
- Add expense tracking, charts and budget analytics.
- Add more planning categories such as travel, education and shopping.
- Improve security with stronger authentication and production deployment.

21. CONCLUSION

PocketSmart AI demonstrates how a web application can combine budget inputs, user preferences, database storage and recommendation logic to support practical planning. The project provides authentication, dashboard, home, party and jewelry planning modules and recommendation history. The successful interface screenshots and HTTP 200 OK server responses demonstrate the working development application. The modular design provides a foundation for future integration with live data sources, stronger AI personalization and additional planning services.

22. BIBLIOGRAPHY

- Python Documentation – Python programming language reference.
- FastAPI Documentation – FastAPI web framework reference.
- Uvicorn Documentation – ASGI server reference.
- HTML5 and CSS3 web development references.
- SQLite / SQL database references.
- Project source code, local database and development test records.

23. VIVA QUESTIONS AND ANSWERS

What is PocketSmart AI?

Answer: It is a web-based smart budget and recommendation assistant.

What is the main purpose of the project?

Answer: To generate budget-aware recommendations according to user preferences.

Which programming language is used?

Answer: Python.

Which backend framework is used?

Answer: FastAPI.

Which server is used for development?

Answer: Uvicorn.

What are the main planners?

Answer: Home Interior Planner, Party Budget Planner and Jewelry Budget Planner.

Why is a database used?

Answer: To store user information and recommendation history.

What does HTTP 200 OK mean?

Answer: The request was successfully processed by the server.

What is the role of the dashboard?

Answer: It provides the logged-in user with an overview of their activity and planners.

What is the future scope?

Answer: Live price comparison, stronger AI personalization, mobile support and additional planning categories.

24. TEAM MEMBERS

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